Transformata Laplace

Def Se numește funcție original Laplace orice funcție $f: \mathbb{R} \to \mathbb{C}$ ce îndeplinește condițiile:

- $f(t) = 0$, $\forall t < 0$
- f e derivabilă pe porțiuni pe $[0, +\infty)$
- $\exists M > 0$ și $s_0 \geq 0$ a.î. $|f(t)| \leq M e^{s_0 t}$, $\forall t \geq 0$

(cond. de creștere exponențială a funcției f, iar s_0 se numește indicele de creștere a fc.)

Mulțimea funcțiilor original Laplace

Se notează cu $\mathcal{O}$ mulțimea funcțiilor original Laplace

Exp: $u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$

$\mathcal{M}(t)$

$\nabla(t)$

Def Fie $f \in \mathcal{O}$ se numește transformarea Laplace a lui f (sau fct. imagine) o fct.

$F: \{p \in \mathbb{C} \mid \operatorname{Re} p \geq s_0\} \to \mathbb{C}$ definită prin

$$F(p) = \int_0^{\infty} f(t) e^{-pt} dt$$

Notăm $f(t) \xrightarrow{\mathcal{L}} F(p) \implies F(p) = \mathcal{L}[f(t)](p)$

Teorema de inversare Mellin - Fourier

Fie $f \in O$ cu s_0 indicele de creștere și $f(t) \xrightarrow{\mathcal{L}} F(p)$

atunci $\forall \alpha > s_0$ arbitrar avem

$$f(t) = \frac{1}{2\pi i} \int_{\alpha - i\infty}^{\alpha + i\infty} F(p) e^{pt} dp$$

pentru orice punct de continuitate t al fct. f

În orice punct de discontinuitate al funcției f, membrul drept este egal cu media limitelor laterale în acel punct

Obs $f(t) = \sum_k \text{Rez}(F(p)e^{pt}, p_k)$

Proprietățile transformatei Laplace

Prop 1 – Liniaritatea

Fie $\alpha, \beta \in \mathbb{C}$

$f(t) \xrightarrow{\mathcal{L}} F(p)$

$g(t) \xrightarrow{\mathcal{L}} G(p)$

atunci $\alpha f(t) + \beta g(t) \xrightarrow{\mathcal{L}} \alpha F(p) + \beta G(t)$

Prop 2 – Asemănarea

Fie $a>0$

$$f(t) \xrightarrow{\mathcal{L}} F(p)$$

atunci $f(at) \xrightarrow{\mathcal{L}} \frac{1}{a}F\left(\frac{p}{a}\right)$

Dem

$$\mathcal{L}[f(at)](p) = \int_0^\infty f(at)e^{-pt}dt = \left| \int_0^\infty f(\tau)\cdot e^{-p\frac{\tau}{a}}\cdot\frac{1}{a}d\tau \right.$$

$at=\tau \Rightarrow t=\frac{\tau}{a} \Rightarrow dt=\frac{1}{a}d\tau$

$$= \frac{1}{a}\int_0^\infty f(\tau)\cdot e^{-\frac{p}{a}\tau}d\tau = \frac{1}{a}\cdot F\left(\frac{p}{a}\right)$$

Prop 3 – Întârzierea

Fie $\tau>0$

$$f(t) \xrightarrow{\mathcal{L}} F(p)$$

atunci $f(t-\tau) \xrightarrow{\mathcal{L}} e^{-p\tau}F(p)$

Dem

$$\mathcal{L}[f(t-\tau)](p) = \int_0^\infty f(t-\tau)e^{-pt}dt = \int_\tau^\infty f(t-\tau)e^{-pt}dt =$$

$f\neq 0 \Rightarrow f(t-\tau)=0 \quad \forall\, t-\tau<0$ $\quad\Big|\; t-\tau=\theta \Rightarrow dt=d\theta \;\Big|$

$$= \int_0^\infty f(\theta)e^{-p(\theta+\tau)}d\theta = e^{-p\tau}\int_0^\infty f(\theta)e^{-p\theta}d\theta = e^{-p\tau}F(p)$$

Prop 4 - Deplasarea

Fie $\lambda \in \mathbb{C}$ și $f(t) \xrightarrow{\mathcal{L}} F(p)$

atunci $e^{-\lambda t} \cdot f(t) \xrightarrow{\mathcal{L}} F(p+\lambda)$

Dem $\mathcal{L}[e^{-\lambda t} f(t)](p) = \int_0^\infty e^{-\lambda t} f(t) e^{-pt} dt$

$= \int_0^\infty f(t) e^{-t(p+\lambda)} dt = F(p+\lambda)$

Prop 5 - Derivata imaginii

Fie $f(t) \xrightarrow{\mathcal{L}} F(p)$

atunci $t^n f(t) \xrightarrow{\mathcal{L}} (-1)^n F^{(n)}(p) \quad \forall n \in \mathbb{N}^*$

Dem

$n=1 \quad F'(p) = \frac{d}{dp}\left(\int_0^\infty f(t) e^{-pt} dt \right) = \int_0^\infty f(t)(-te^{-pt}) dt$

$= \int_0^\infty (-tf(t)) e^{-pt} dt = \mathcal{L}[-tf(t)](p) \Rightarrow -tf(t) \xrightarrow{\mathcal{L}} F'(p)$

$\Rightarrow tf(t) \xrightarrow{\mathcal{L}} -F'(p)$

Pp adev. pt n și demonstrăm pt $n+1$

$\mathcal{L}[t^{n+1} f(t)](p) = \mathcal{L}[t \cdot t^n f(t)](p) = -\mathcal{L}'[t^n f(t)](p)$

$= -[(-1)^n \cdot F^{(n)}(p)]' = (-1)^{n+1} \cdot F^{(n+1)}(p)$

Prop 6 – Derivata originalului

Fie $f \in O$ cu $f(t) \xrightarrow{\mathcal{L}} F(p)$

Pp. $\exists f', f'', \dots f^{(n)}$ cu $f^{(n)} \in O$

$$f^{(k)}(0_+) = \lim_{\substack{t \to 0 \\ t>0}} f^{(k)}(t) \qquad k = \overline{0, n-1}$$

atunci $f^{(n)}(t) \xrightarrow{\mathcal{L}} p^n F(p) - p^{n-1} f(0_+)$

$$- p^{n-2} f'(0_+) - \dots - f^{(n-1)}(0_+)$$

Dem

$$\mathcal{L}[f'(t)](p) = \int_0^\infty f'(t) \cdot e^{-pt} dt = f(t) e^{-pt} \Big|_0^\infty + p \int_0^\infty f(t) e^{-pt} dt$$

$$|f(t) e^{-pt}| \le M e^{s_0 t} |e^{-pt}| = M e^{s_0 t} |e^{-(\sigma + i\eta)t}| =$$

$p = \sigma + i\eta \quad \sigma = \operatorname{Re} p > s_0$

$$= M e^{s_0 t} |e^{-\sigma t} e^{-i\eta t}| = M e^{s_0 t} e^{-\sigma t} = M e^{-t(\sigma - s_0)}$$

$$\Rightarrow \mathcal{L}[f'(t)](p) = -f(0_+) + p \mathcal{L}[f(t)](p) \quad | \cdot p^{n-1}$$

În egalitatea de mai sus se înlocuiește f cu $f', f'' \dots f^{(n-1)}$

$$\mathcal{L}[f''(t)](p) = -f'(0_+) + p \mathcal{L}[f'(t)](p) \quad | \cdot p^{n-2}$$

$$\mathcal{L}[f'''(t)](p) = -f''(0_+) + p \mathcal{L}[f''(t)](p) \quad | \cdot p^{n-3}$$

$\dots\dots$

$$\mathcal{L}[f^{(n)}(t)](p) = -f^{(n-1)}(0_+) + p \mathcal{L}[f^{(n-1)}(t)](p) \quad | \cdot p^0$$

$$\oplus\ p^{u-1}\cdot\mathcal{L}[f'(t)](p)+p^{u-2}\mathcal{L}[f''(t)](p)+p^{u-3}\mathcal{L}[f'''(t)](p)+$$

$$+\ldots+\mathcal{L}[f^{(u)}(t)](p)=-p^{u-1}f(0_+)+p^u\cdot\mathcal{L}[f(t)](p)-p^{u-2}f'(0_+)$$

$$+p^{u-1}\mathcal{L}[f'(t)](p)-p^{u-3}f''(0_+)+p^{u-2}\mathcal{L}[f''(t)](p)-$$

$$-\ldots-f^{(u-1)}(0_+)+p\mathcal{L}[f^{(u-1)}(t)](p)\Rightarrow$$

$$\Rightarrow\mathcal{L}[f^{(u)}(t)](p)=p^u\cdot\mathcal{L}[f(t)](p)-p^{u-1}f(0_+)-p^{u-2}f'(0_+)$$

$$-p^{u-3}f''(0_+)-\ldots-f^{(u-1)}(0_+)$$

Prop 7 – Integrarea originalului

Fie $f(t)\xrightarrow{\mathcal{L}}F(p)$

atunci $\int_0^t f(\tau)\,d\tau\xrightarrow{\mathcal{L}}\frac{F(p)}{p}$

Dem Notăm $\int_0^t f(\tau)\,d\tau=g(t)\Rightarrow f=g'=pt$

$$F(p)=\mathcal{L}[f(t)](p)=\mathcal{L}[g'(t)](p)\overset{\text{deriv. orig.}}{=}p\mathcal{L}[g(t)](p)-g(0_+)$$

$$\Rightarrow\mathcal{L}[g(t)](p)=\mathcal{L}\left[\int_0^t f(\tau)\,d\tau\right](p)=\frac{F(p)}{p}$$

Prop 8 – Integrarea imaginii

Fie $f(t)\xrightarrow{\mathcal{L}}F(p)$

$\frac{f(t)}{t}\in O$

atunci $\frac{f(t)}{t}\xrightarrow{\mathcal{L}}\int_p^\infty F(\theta)\,d\theta$

Dem Notăm $\frac{f(t)}{t} = g(t)$

$$g(t) \xrightarrow{\mathcal{L}} G(p)$$

$$f(t) = t \cdot g(t) \Rightarrow F(p) = \mathcal{L}[f(t)](p) = \mathcal{L}[tg(t)](p) \overset{\text{deriv. imag.}}{=} -G'(p) \Rightarrow \int_p^\infty F(v)dv = -\int_p^\infty G'(v)dv = -G(\infty) + G(p) = G(p)$$

$$= \mathcal{L}\left[\frac{f(t)}{t}\right](p)$$

$$|G(p)| = \left|\int_0^\infty g(t)e^{-pt}dt\right| \leq \int_0^\infty |g(t)e^{-pt}|\,dt$$

Notăm $p = \sigma + i\eta$

$\sigma = \mathrm{Re}\, p > s_0$

$$\leq \int_0^\infty Me^{s_0 t}\cdot|e^{-\sigma t}e^{-i\eta t}|\,dt = M\int_0^\infty e^{s_0 t}e^{-\sigma t}dt$$

$$= M\int_0^\infty e^{-t(\sigma - s_0)}dt = M\cdot\frac{e^{-t(\sigma-s_0)}}{-(\sigma-s_0)}\Big|_0^\infty = \frac{M}{\sigma - s_0} \xrightarrow{\sigma\to\infty} 0$$

$\Rightarrow \lim_{\mathrm{Re}\, p\to\infty} G(p) = 0$

Prop 9 – Convolutia

Fie $f(t) \xrightarrow{\mathcal{L}} F(p)$

$g(t) \xrightarrow{\mathcal{L}} G(p)$

$$(f*g)(t) = \int_0^t f(\tau)g(t-\tau)d\tau$$

(se numește produsul de convoluție dintre f și g)

atunci $(f*g)(t) \xrightarrow{\mathcal{L}} F(p)\cdot G(p)$

Dem $F(p)=\int_0^\infty f(\tau)e^{-p\tau}d\tau \Big| \cdot G(p)$

$$F(p)\cdot G(p)=\int_0^\infty f(\tau)e^{-p\tau}G(p)=\int_0^\infty f(\tau)\left(\int_0^\infty g(t-\tau)e^{-pt}dt\right)d\tau=$$

$$\| \mathcal{L}[g(t-\tau)](p)$$

$$=\int_0^\infty\left(\underbrace{\int_0^\infty f(\tau)g(t-\tau)d\tau}_{(f*g)(t)}\right)e^{-pt}dt=\int_0^\infty (f*g)(t)e^{-pt}dt$$

$$=\mathcal{L}[(f*g)(t)](p)$$

$g(t-\tau)=0 \quad \forall\, t<\tau$

$f(t)$	$F(p)$
$u(t)$	$\frac{1}{p}$
t^u	$\frac{\Gamma(u+1)}{p^{u+1}}$
$e^{\omega t}$	$\frac{1}{p-\omega}$
$\sin(\omega t)$	$\frac{\omega}{p^2+\omega^2}$
$\cos(\omega t)$	$\frac{p}{p^2+\omega^2}$
$\mathrm{sh}(\omega t)$	$\frac{\omega}{p^2-\omega^2}$
$\mathrm{ch}(\omega t)$	$\frac{p}{p^2-\omega^2}$

$\Gamma(u+1)=u!$

Obs

Toate funcțiile care apar în tabel se presupun a fi înmulțite cu $u(t)$